BGIN Block#11 Meeting Report

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CryptoISACs with BGIN

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BGIN Block#11

CryptoISACs with BGIN

Date: Monday, October 22, 2024

Location: Washington DC, United States

Duration: Approx. 90 minutes

Summary

Session Overview: The second session today is focused on Crypto ISACs (Information Sharing and Analysis Centers) with BGIN, aimed at building a foundational understanding of cybersecurity for the blockchain ecosystem. This session will assess the effectiveness of information-sharing efforts, highlight challenges, and discuss how to quantify and improve collaboration among stakeholders.

Tokyo and Sydney Discussions: We presented this initiative, engaging a community that included Bitcoin developers and members of ISO TC307. Key points discussed:

- **Threat Intelligence**: The blockchain industry lacks maturity in sharing specific intelligence items. Adapting frameworks from other industries may help create a tailored list for crypto.
- **Taxonomy**: We need a well-defined taxonomy for threat categorization specific to the crypto industry.
- **Timing of Information Sharing**: Developers expressed concerns about sharing vulnerabilities prematurely, especially if they remain unexploited.
- **Qualified Experts**: Many projects lack sufficient cybersecurity expertise, limiting the utility of shared information.

Country-Specific Concerns: Participants discussed the nuances of sharing vulnerabilities with governments. For example, Bitcoin developers noted that sharing protocol vulnerabilities with El Salvador's government might be unproductive if expertise is lacking.

Decentralization and Traditional ISACs: Decentralization in DeFi and blockchain projects presents unique challenges. Collaboration with traditional ISAC frameworks, which focus on users of specific technology, requires adapting those frameworks to the blockchain industry.

Regional Collaboration Needs: Due to the international nature of crypto, a country-by-country approach is less effective. Collaboration on a global scale is necessary.

Japanese Perspective on Crypto Security: Japan’s Crypto ISAC initiative shared insights:

- The crypto industry in Japan faces frequent security incidents, particularly with cold wallets. Incidents reveal that even "secure" cold storage can be compromised.
- Japan's Crypto ISAC, called JP Crypto ISAC, aims to strengthen industry resilience by bringing together stakeholders and implementing robust security measures.

Challenges Faced by Crypto Companies in Japan: Key challenges include:

- **Transaction Sensitivity**: Transaction data is sensitive and often handled inconsistently across companies.
- **Operational Security**: Secure key management remains a unique challenge in Japan's crypto sector.
- **Limited Resources**: Small companies struggle to hire adequate security experts due to budget constraints.

Standardization and Collaboration: The discussion turned toward the importance of international standards for cybersecurity:

- **Existing Standards**: C4 in Canada has developed a key management standard. The Blockchain Security Standards Council is also working on cybersecurity standards for crypto.
- **Incentivizing Compliance**: Compliance can be driven by regulatory pressure, insurance requirements, and internal resilience goals. Having stakeholders like venture capital firms involved can reinforce the adoption of standards early in a company's lifecycle.

Workforce Development and Skill Gaps: There is a recognized skills gap in crypto security. Initiatives like CyberSmart aim to address this by collaborating with educational institutions to build a skilled workforce for cybersecurity needs.

International Standards and Decentralized Compliance: Different approaches to standards exist between traditional industries and blockchain. Blockchain's decentralized nature requires standards that accommodate diverse stakeholders, which could be driven by organizations like BGIN.

Proposals for Standardization in the Crypto Sector: Proposed strategies include:

- **BGIN as an Ecosystem Orchestrator**: Acting as a central body to foster cooperation, education, and common standards without enforcing strict compliance.
- **Educational Resources for Global Reach**: Developing guidelines for regions without established ISACs to help them create cybersecurity infrastructures.

Roadmap and Next Steps: Post-Block 11, BGIN plans to establish a cybersecurity working group. This group will develop educational content, work on information-sharing alignments,

and explore collaboration with ISO for potential global standards. BGIN invites all interested members to participate.

Conclusion: The session emphasized the need for a collaborative, standardized approach to tackle cybersecurity challenges in the crypto industry. Upcoming initiatives by BGIN will help create a cohesive framework for international cooperation on crypto security.

Transcript

That's the second session of today is the Crypt ISACs with BGIN and the way for the hub the very very very critical discussion in the process session. And so we have that common view or commander standing on that entire picture of cyber security for blockchain ecosystem. And so what's that so that information sharing is working or it's not working and what is a potential challenge so you can also sharing activities then so after having that kind of that conversation for that that quantifting so we double greek that discussion to identify at what kind of information should be shared when or something like that. can do and cannot do and how big he can do facilitate that so that that's full to full cooperation among stakeholders.

And so we presented that work we discussed in Tokyo with a vital core community and in siderly I saw TC307 community one that crypt is that so even if you're sharing things And after having that and feedback from those communities So I would ask could i look like JP? persons to present so what they are doing now anyway so let's get started with the feedback from type should we start this presentation before that meeting.

Okay, so, okay. I don't know where to stand, but… I don't know. A couple of beef. Yeah, she'll be real good. Two weeks, yeah, right back, right? Bye. Listen. Okay. Yeah. Now that you have… Okay, let's start the presentation so And as some of you have already known it. We talked about the new thing the cybersecurity International Sharing alignment stuff In May, to meet up in Tokyo in sydney and also we talked about it in San Francisco as well so we would like to talk about discussion point we made in introduced tokyo and Sydney in conferences we had and um It's very easy to put on the screen But yeah, here is an overview of what we discussed in Tokyo and um technique.

We already talked a lot about like threat intelligence in the last session, but in Tokyo event still a lot of like, you know, there were some Bitcoin plot developers in that event And we talked about the industry immaturity In terms of like certain intelligence. So what I mean by saying that is that you know the blockchain and crypto industry hasn't matured enough to identify specific items to be shared and knowledge and it's shared or exchanged they said but that's why you know um we needed to start by looking at existing ISAC framework in other industries. We can create a new list we can create a new lift for crypto by modifying those existing lists and adapting them for the crypto industry, which is one of the discussion point we made.

And also we need to come up with a taxonomy for certain patients as all the participants said. the ones redefining a taxonomy for the crypto industry, we can think about the data model and we talked a lot about in the last session, but it would be great to have a list of more mature states documented and this is also future work item. and also some participants talking about the timing of the information sharing So when we talked about it with Bitcoin developers, they say that they will not be there will not be any resistance to sharing information about funds But when to share about all the information you feel in the session but in some cases, people find bugs that are not yet exploited So you don't want to immediately disclose them.

On the other hand, vulnerabilities like the DOS issue in Bitcoin were fixed years ago, but only disclosed later to ensure that everyone had upgraded. right so timing sensitive information is always exists so when to share those information is very important as well. And so there was also discussion around the expertise of ISACs and also vendor companies like even if it's a good scheme to share information, if each company or project doesn't have an expert in advance the information um It's not probably helpful so it's either should be is very good good scheme. Another issue is whether each organization or company or project has enough qualified experts so this is also another discussion point to be made.

And also um because there was also an opinion about the wild there was no problem with sharing the information about the bowling abilities of the like you know For example, Bitcoin protocol with the government so there was a person who is from El Salvador in that session and they said that um So the Bitcoin core developer said. something said to the ill foldable person union from El Salvador and there was an opinion like while there was no problem with sharing information about the vulnerabilities of the Bitcoin protocol with the government of El Salvadol. They do not share this information because they do not expect the government to understand everything. But they said it's okay to share those information but it's okay So they just do not do it because you know they do not expect that they understand everything so And… also decentralization is also another point.

That point was also made in the last discussion as well but most foundations and projects in DeFi claim that they use decentralized technologies with no central entity ripples too so it is also another problem that we already discussed in the last session. so also in terms of collaboration with traditional ISACs The traditional ISAC framework is based on users of specific technology sharing information. So we need to consider what part of blockchain services can be categorized as an industry so that we can find the common merits or outcomes protected by sharing information. And also the thing that we already talked about in the last episode.

yeah and um yeah also in terms of correlation across crypto ISACs in the US, due to the international nature of crypto assets country by country basis is not effective. So international collaboration systems necessarily is one of the points we made in that discussion in Tokyo and

Sydney. So this is a quick overview of what we discussed, and I think it's a good transition to And next. But the question we have in this question on agenda.

Thank you very much for sharing what we discussed so far on this topic with as a community then Is there a question about that? Okay, so… So as you see that at the end of this session so that's a frost proud of this session so we would like to identify that current challenges for existing ISACs. So we have very big entire picture for the cybersecurity blockchain world you know cook daisak is a very important stakeholder you know that that information sharing is the foundation of that to make the every every everyone uh is very pretty be prepared to secure everything but at the same time that crypto is actually Whoa Isaac has had so many talents to keep with that you know you know sustainable operation So it elevated studies and they're supposed to criticizers so So that might be additional challenges than that pick 16.

Isaacs. Anyway, so at the first part of this session So I would like to, because of many discussion about that. So to identify that kind of the challenges about crypticex. And at the first, so I would like to ask sasakisan So he's a part of the newly developed in cryptocs in japan to give a presentation at what or doing australian now.

Thank you. Hello, let me share the situation about cryptocurrency in Japan. So here is the security instance or create asset provided in Japan. So there are many many instances. So also, cryptocurrency companies use crime to keep security, security risk at various variations but unfortunately the incident happened many times.

So if you ask the company to keep the loose and guidance. So after that, we have tried to keep uh security manner rule and so on. But unfortunately, the incident happened again. So probably, as you know so that latest incident happened in both or sorry trash in this trash. So this was a really big incident and So as you see, the situation happened in the Cold Wallet. So in Japan, so Japan's cryptocurrency company No, the cold pork is really safe. And the issue don't happen but not so this issue happened.

So why this happen? But surgery, the country price is just investigating what happened. But I would say we don't know what happened and what happened and so on. So for me, so about two years ago the proven loop had a contact pick in Japanese cryptocurrency company. This is the one that we do company to do it. So at that time So because the primary loop it was not he's like, so a lot of credit cards companies asked me uh to read So as a transfer, I tried to gather a prophetic company and we i made a new uh to keep the probability good.

so and probably very good have finished but recently as you see this issue happened so this was a reason why a lot of members asked me to this the jp uh crypto ISACs. So this is the one piece. So let me share the challenges facing the cryptocurrency industry. So this is just the prospecting of the cryptocurrency company So one cost is a transaction size handled by each cryptocurrency

company. And this is a really sensitive information because we don't know as a personal capacity a company do it Second is operational using secret key is very, I think, unique.

is in the suite because so we have to operate by using a secret key. by Banya because this is really important key factor. Probably these were the most reason why this industry is already targeted Because, so after speaking after so steering the ticker key. mini currency. we distribute was distributed many things and after that we cannot so probably uh check or monitor this and we can Pirate, give up. so but uh so crypt friends company uh have to keep the asset of customers. This is a really important thing because a customer rely on us.

So this is why we have started to create JP Crypto ISAC So let me share this. The name is Japan crypto ISAC. So we are just breeding this conversation and probably this organization will be um created by the end of this year. December, for example two days ago, Japanese nikkei introduced this organization And the purpose of it here and also featured it be here. So this was the reason why i asked Dr. How to proceed it. And I have just heard that in us the filter is already free it and they try a lot of things So I decided to cooperate if possible. So these are the reasons why I go to here and I pretend here So that's three. So international connections is a really important thing because put a currency is uh not in Japan, in the world. And we have to keep and we have to keep this industry because this in Japan. Also, we try to do uh sometimes it happens And at that time customer Always really not satisfied.

So this is just situational. So that's it for me. Thank you very much about that. Any question about that situation in Japan? But that project is not so for the presentation, but he'll hear that you know it's a Chatama household recession He can speak up. But any question about that conversation in Japan?

So they have members She's speaking. Members. So I am in charge of the crypto uh this organization? No. Like industrial numbers. So as part of them, that group data today really is now starting to recruiting.

Okay, so I would like to ask some questions about that because that's the goal of that path forward is identifying those challenges you know so you know that you know that after the incident happened that this july this year so you know many people requested to initiate that so So I'm very curious about so what you know they say include you know japanese politicians or japanese government So who tries to request to cryptize activity to conduct That's kind of equal you know what is the goal of the crypt isaac.

Yeah, so the prior work is to keep, as I explained, to keep the customer's asset safely on the big codes so each try to keep Saylor, customers opted. But sometimes it should happen. so they skip Why have them but they don't know. in terms of prosecutive marriage and so on. So they probably sync to create powerful institution, a powerful organization it's important thing to collaborate each other is really important thing also the secret is what narrative that really tells

you but we have to beta check and if something happens, we have a better share That sounds possible. So this is the reason.

Thank you. I'm curious about within your prospective member companies what sort of maturity you see internally within those companies when it comes to information security. Do you think those member companies already have threat intelligence capabilities internally? Or is that something that you think you might have to help the member's bills. And who would be the primary sponsor, do you think? that would lead the membership.

So these are just online, but currently each company hire the security members that they tried to gather each other by themselves But I want you to think, this depends on the company's size and so on. So because in japan there are really small companies But unfortunately, due to the lack of budget and so on, they cannot hire security Me and Brad are so warm. but they have to keep custom of assets yeah this is.

Is there, you know, here is a requirement for public companies to identify responsible parties for information security. In other words, there should be a CISO. Is there any regulatory requirement in Japan for that kind of governance model?

Yeah, so if you say ask us to keep to keep the customers at it and they pretty to the blue. But I want you to read it. I think this is a ridiculous. bit of that their information because probably a This is just us. here but i think As I explained that companies there are many big large middle small companies and they don't know how they proceed.

But do you think the monitor? Many people have to be to read it. Although I am not security engineer, but they asked me to proceed to cooperate with your chi and you and So I think I did say the important point. So what I asked the previous session about that workforce development or a member of that experience and qualified experts You know, so it's it.

but there's three different types, you know, like a cso so so they are responsible for the you know that organisers herself. And then, you know, that security engineers who picks us a program or the network or protot, something like that and operators So Z, different types of you know expert needs so that company needed So this is a sole I'm usually talking about so it's about the difficulties the quick company can has to build. cannot hire that, you know, corporate number of experts you know that for example that you know

So there are many, so he said happened in japan so and then i don't know about that. So Samuels might be initiated by some nations, for example. So kind of a cyber war So abominations. Potentially. But, you know, great company need to protect their system against a national level That is it.

a huge headache. And then, you know, at the same time, most of those companies are very small startups So they don't have any qualified budget to hire that qualified number of people.

The other thing is, you know, there's no qualified number of export interval to handle that nation-level protest. Yeah, so it's a so it's the same as a wall. So they need to have long time expense, one year, two years, three years experience to fight against the nation in cyberspace. It's a problem of experience. So we don't have, we cannot, so that kind of person in one day or two days. So we need three years to have the experienced people. So even if we have tornial money or money, but so we cannot have that, that kind of problem.

These countries that we can bring that person from the, you know, for the internet service provider or something like that, they have that on the corporate expert group have a good experience fight against some nations. But then, so it's easy to educate some blockchain things for the expert in the ISP. So it's easy to educate with the blockchain technology after two weeks or three weeks. But it's not easy to educate the blockchain guys in cybersecurity. It takes three years. That kind of thing is a source of a headache.

And of course, they are hired by the infrastructure company that so there are a lot of people. You know, fundamental system in the world, and we cannot bring those systems and quick world, so that is a source of a headache. And I think that that kind of problem also there in the United States or other countries. I think, you know, I don't know if this is working.

You know, we spent all these years talking about the cybersecurity skills gap and then we've got all this investment into educating a whole sort of workforce. There's definitely a crypto security skills gap, 100%. And um, I've witnessed it over the past several months. You've got security experts, you've got crypto experts, there's not a whole lot of overlap. And there's people that have sort of spanned both are very, very busy. So I think there's definitely an opportunity to address education to your point.

And of course, that, for instance, that's working on that workforce development for cybersecurity. You know, some crops, you know these are not so easy but they're cooperating with universities or, you know, like a partner tech is some essential part. Thank you. They have something to speak about. Okay.

Yeah, so we have Virginia Tech and actually across the Commonwealth, the accent that's the Commonwealth of Virginia, not the Commonwealth. Well, that's right, from the times when we were part of the Commonwealth. Bye. Where it is all the public universities and colleges. Because workforce development is not only about producing PhDs who write papers that no one reads that gets them tenure track appointments at high output PhD granting institutions where they create more PhDs who write papers that no one reads.

Yes, we do invent our way out of a lot of this. One of the points of sticks was to be able to find patterns that humans might miss. So that's a PhD level stuff, but you know, you even need someone actually looking at after the AI picks out the interesting alarms, you still need somebody to do that. And that could be someone who's just got a lot of experience and an associate's degree. I hear the army and stuff like that, right? So, you know, hitting all the areas and a lot of the problems we see.

Is companies like I can see it in the crypto industry you know we want someone with five years of experience in cybersecurity, 20 years of experience, and I picked that number judiciously in blockchain technology. And, you know, three years of management experience. Well, the reason I picked that number is, you know, the Satoshi white paper was 2008. Nobody, literally nobody, has 20 years of crypto industry experience because it was before the.

Yes, the pieces were there. That's right. That's right. He's got a good time machine. So we also have programs, you know, besides all the research and interesting stuff, consulting that people do. It's like we have one program where for companies in Virginia. So we do help. We're confident enough in the program to be exported to other states, where a professor will work with two or three students so students go into a small business.

Or municipal government because we're talking. A lot of these companies are really small. You know, we've had startups with, you know just a handful of people. Well, this is a way where the students can work with them to raise their cybersecurity posture. And they're getting that experience. And because we don't believe in slave labor, they're actually paid for by the state. We actually pay. And at the end of the day, the company now has a better cybersecurity posture, students now have experience so they can get a job and repeat.

So a lot of stuff going on in workforce. Can I skip something? I didn't use my cyber smoke. I forgot to mention that Cypress Smart was actually started by Georgetown in 2019, yeah some guy sitting here at Georgetown. And then they joined in 2021. Virginia Tech joined just this summer. And so we are also in the process of bringing in the ocean. It's a civic it's a technology university of Sydney.

And they'll be joining probably very early next year. But also how we work with industry is we create, it's all project driven and for each project we would hire probably three, four students who could solve, you know, changing problems that are of interest for the industry member. And this is a workforce development program, right? Train new cyber for cyber workforce. And these students will graduate, they very often are getting hired by this company.

So again, if you're looking for new members, if anyone's interested. And something that's uh program. If you have a project that requires deep expertise in cryptocurrencies, deep expertise in forensics, and deep expertise in cyber, it's all set up so that we work together. Which is

something that's kind of rare in academia. So it's kind of rare in industry too. The program is 40 years old.

And so they've worked out all the electric property things so that multiple companies can work with multiple professors at multiple institutions with students from different places to get things done without triggering any trust. So now that we've totally hijacked your session. Yeah, hijacked that. So by advertising it. Brought to you by…

Okay, anyway that's you know that fourth point is education and workforce everything that that is a yes. Yeah, so they're one of the headache or the category is now that the other I would like to go here that opium from or you want to use the standards. No that you don't. So diagrams showed diagrams. It's a previous session is related to that standard.

Responsible disclosure protocol itself is standardized at ISOIEC. That is also, you know, gold standard in the cybersecurity space standardized process, something like that. So we don't have to be denting oil. So we should deliberate that existing framework and we can add something to handle the crypt specific topic. At the same time that's so.

So I will be at the mini discussions always equipped industry guides. But it's related to the educational thing but then, you know, crypto guys are very young people that they don't have enough knowledge about the existing cybersecurity framework for, you know, standard they sell. They don't care about that. So they copy and paste that something called from original services page so call and tweak something like that to bring that to the to the real world.

But they don't care about that standard. I was a cryptography and protocols, you know something like that or cybersecurity, but they don't care about that. So that program is itself, if we can provide some information to every young people in the world in this world, so that could be good but then. So this is another thing, another story so.

You know for the interns. So we will discuss about the interns session for the internet so standardly it's important i even also making the standard because stand out in the market for the you know cryptocurrency. So they can earn some money, some amount of money without the standard. So they don't have any so elaborate, you know incentive to make standard.

That is a defense against the internet ecosystem. But anyway, but for cybersecurity standard that you want to. But I think that so most of people that don't care about the standard at this moment. So I'm going to ask Jesse first after you have a conversation with companies or with young people. So what is the clinical status of that awareness of the cybersecurity standards? Yeah, sure.

Yeah, so definitely standards are a necessity. And we, you know, we are aware of a few pre-existing efforts in the standard space. All these vulnerabilities here, these hot wallet and cold

wallet key management vulnerabilities were tackled by a one standard that I think is already out there by an organization called C4 out of Canada. They have a standard.

I don't know that I understand it. I'm sorry, but it's related to key management for crypto. There's an emerging organization called the Blockchain Security Standards Council that's also a representation of industry that is also working on standards for cybersecurity and crypto. In this begin and other organizations starting to tackle standards.

So the crypto ISAC is collaborating with these organizations to ensure that these standards, the programs that we develop align with these standards. Another point I was going to make is how do we incentivize stakeholders to comply to meet these standards? What's in it for them, right? And we know that there are a few um, you know area there are a few.

Types of organizations that can incentivize and motivate stakeholders to meet these standards. One is the regulators right. Regulatory pressure will require compliance. And one way to demonstrate compliance is to meet standards. So that's one thing. There's the insurance industry. Insurance the insurers may choose to issue policies that are not based on stakeholders attesting to the fact that they meet these standards.

And of course, there's the audit community that goes along with that. And then this is best practice in doing the right thing and building resilient and secure infrastructure, which we would hope that everybody wants to do. There's a reason why the crypto ISAC has such a diverse set of stakeholders, including venture capital in its founding membership.

That is because these investors and folks in the investment community want to ensure that their portfolio companies who can be at a very early stage consider and fold in security best practices with a goal of meeting the standards that are out there or that are under development as they grow.

So this kind of comes back to us recognizing, I think, the diversity in crypto in the community. Including folks on the government side, on the regulatory landscape and the insurance industry, which also, by the way, is represented in the crypto ISAC. And within the investment community.

Again, there's a reason why electric capital ribbit Capital, sits on our board of directors. I mean, we are as welcoming to the folks from the investment community as we are within the crypto infrastructure and solutions provider landscape.

Thank you very much.

Is there any other standard? You know I'm not saying that I feel uh doing so many discussion a couple of years. The definition of standard is different among the industry. You know that for the

internet industry that somehow is a subject, IETF standard or something like that. It's a community-based standard.

And no one prohibited to use some technology outside of the IETF study. So cut that so. It's a discussion topic of the last session but then so okay it might be a question to Keralos otherwise so. How can we create that standard with some enforcement powers. For cybersecurity data to you.

Oh.

Well, yeah, and I think that our standards bodies would also love to highlight that they don't necessarily love creating standards under the context of them being enforced against. Of course, we know that regulators and others absolutely lean on them. I just know standards bodies often are like, no, we're not regulators. I'm like, I know exactly how regulators are going to use your standards in their regulations. So it's interesting, like the dynamic between the two communities is interesting and necessary since they have different functions.

To me, I think first, you want to create a standard that will have adoption, so that's attractive to the industry, which means looking for something that is technically sound, practically implementable, meets the objective of it. And then I guess something that has broader stakeholder buy-in. I know that's the point of this group, but like something that I think will be a major challenge that is especially unique in the crypto industry.

Is how decentralized the community is. Decentralized communities have a hard time organizing. It took a long time for them to create the ISACs. It took about five years from when we first started calling for it. Thrilled that they're happening and already operational. But it took time. The traditional industry also a long time to create these things. So that's not that's not just unique there. I just do wonder like how you get.

How you get a decentralized community to more broadly adopt something, but basically that's going to be a consideration that regulators take into account when thinking about enforceability of things. Many nations have not yet enforced against the FATF travel rule because they claim that since there's no technology that was built to facilitate its compliance, that they shouldn't enforce against it.

I personally take issue with that characterization. Nothing in the travel rules states how it has to be transmitted just because industry decided to create this new bespoke system and is willing to facilitate financial transfers without knowing who's on the other side of it. I don't know how you comply with sanctions, but so be it.

But that was an example of where many nations have chosen not to enforce a standard because they feel that industry has not yet risen to create the capability and the capability in a market

where there's enough like competition and that it's widely accessible, et cetera. So that's like, it's just an example that I would point to. For me, though, I think that like demonstrating that it's technologically feasible and doable will mean that certainly more mature.

Regulators and ones that are more interested in actually enforcing it. I think in combating illicit finance, this is a really good one. If you're talking about other types of standards, like in the prudential world where we've not all agreed to, like, I think even cybersecurity standards is going to be tougher than.

Like then i think even cyber is going to be tougher than anti-money laundering related standards happily for us, the cyber standards can be very much pointed to countering the illicit finance. So I think that we can make more headway there. But AML, just because the international policy standards have existed for longer, you have more maturity in the space to agree around what's needed. So I think that.

There's a variety of factors, I guess, that need to make the opportunity ripe to making that movement. But then once you have regulators seeing that, okay, it's technologically feasible. Stakeholders all see that this meets that objective, then they know that there's something to enforce against. So I guess.

Creating the standard that's technologically doable, which in this case we know it totally is because Sticks and Taxi is a great example. We know that you can share machine readable digital indicators with each other. That gives regulators a good excuse to go and hammer an industry when they've chosen not yet to implement necessary things. So I think that that's part of what you need. But I'm sure you have a lot of views.

This is dull for me.

Go ahead.

Oh, so I've been I've been a security minion for a long time. One conclusion that I came to is the only way to somewhat prove that you have effective security controls in place is to be audited.

And there are no audit frameworks tailored for crypto today, if I'm not mistaken. So I believe it will be nice to have some sort of audit framework. Nobody likes audits, but might need that.

And what I also think is like if we could publish things as an ISO. I was thinking of like something like common criteria, which is often a requirement or, you know providing solutions to the government didn't want to match.

Yeah, so there's standards, standards and standards. You mentioned standards AML, which come from Basel two three four, which basically had the force of international law about people on the

street will say there's no such thing as international law they're just laws that everyone agrees to implement.

And then there's standards like sticks and taxi you know i can download it, read the directions like that open source repositories, I can build it. And then there are standards of behavior.

So I did read something. Especially if you've been setting background, you don't get to answer this. Who thinks TCID DSS is a regulation?

Because looking at the minutes, it said beat block 10, should we have regulations like PCI DSS? This CCIDS as a regulation.

No, it's not it's uh voluntary because, you know, Frank gave us the history of visa it's voluntary because if you want to process credit card charges, Visa makes you do it. It's a commercial agreement. And the banks you process, you know do the visa payments, they signed the contract saying, yep, we'll be PCI BSS. And why because.

It's a common framework that people say, I do this, it's done. Now, as we heard online as having been having chair of an audit committee on a company board, there are audits and assessments.

And this is a space where there's pretty much only assessments. Do you have good cybersecurity? Policies. Can you survive the penetration of this?

Audits are are you DSS compliant? Are you SOC compliant? The weird one is, are you in this framework compliant? That means.

And then the thing I had to laugh about when we were putting together the NIST framework. In the 2011 to 2014 timeframe, this was their going, oh, we're this we're not a regulator. No, no, no, we're not we're not. And like, especially the energy industry in particular was like.

Yeah, DOE is going to make us do this, aren't we? Guess what happened? Careful of what you ask for you. And it was quite a standard, you know. As far as I say, no doubt. That was almost… Just a thought. So as part of that, I understand. I understand that or ITU stand that the international standard and you know under the WTO TVT algorithm the government should the photo that's standard. Procurement, I think.

Sure. So… It kind of goes the other way. Many countries will only buy things that have an international standard behind them, like I said to you. U.S. government is relatively not that because the IETF is actually, now it kind of sort of is, but the IETF is not a standards body, which means TCP IP, which I think the US government buys a little bit.

Would not be able to keep on.

This for the federal civilian, NSA for uh. Bedroom or for U.S military you know they do set standards that the government has to follow, but that's because the president can say. You will only buy stuff that's misapproved or an SA approved. They think, you know.

In this country, the government does not say. Oh jp morgan chase. You have to buy something like this. And they really there's no way they can say, hey, little startup, you know decentralized crypto startup with two people, the US, four people in estonia and. You know a person in South Africa, you have to follow. That's just not the course. Not quite practical I think it was.

What are both of you excited? Is that he has a… There you go benefit from standard with the whole. Having core understanding. So by different different types of stakeholders it's a kind of education. You know that that process of standardization of something is a you know. Footage that documented sale. By using the different types of hands or something like that. But it's a kind of education.

I guess one of the questions that I have when we're speaking about these standards is, the crypto industry is unique in the way that there are different stakeholders within the infrastructure as well, which tend to be separate. So if I'm running a node for a particular network, I may not even have—I'm kind of just a person who runs nodes. I'm not specifically associated with the Ethereum Foundation or specifically associated with NEAR or Solana or whatever. So there isn't really like a top-down incentive for these foundations to put standards on that other than the security of the network, which is fantastic.

One idea that's come to mind is, are we focusing these standards on the sort of centralized app/service providers of the industry who use the assets and the technology in order to support retail or institutional investors, whether or not it's financial or non-financial? Is that the target of the standards? So then we can actually have an intra-organization cybersecurity policy and framework, which is executable and at a high level. Or are we interested in sort of tackling the decentralized infrastructure side of the market, which, to me, is a more complex question but also an opportunity because of the way these organizations operate in the public and in the open. A lot of them have proposals and these sort of different frameworks that allow you to get things in front of the people in the network. Could there be an opportunity for crypto ISAC or one of these organizations to just start writing proposals to DAOs and be like, "Hey, we have this expertise, we have this coordination between cyber smart. Do you want access to it? Do you want our knowledge? Do you want our information? Do you want capacity building within your sort of decentralized collective?" Then just leave it as a note in the organization. You might find that when a vulnerability pops up and some news happens, people will go back to that and be like, "Oh, we actually had an opportunity to mitigate this issue." You get a few of those over the line, and then you start. This industry runs on attention in a really weird way as well. If you get enough attention to that, people will start coming out of the woodwork and asking for this service. Just a few thoughts on that.

One of the issues that we brought up about the standard for it to be enforceable is for it to be agreed upon and something that people will actually use. We know that the ISAC will be interested in using it because they've committed to it. They've talked about it as part of the whole business model. Like, they totally want this. Do you guys feel that this is something that you would use and would be fine or arbitrary or okay to use? Is there some competitive advantage that you guys feel might drive you not to be terribly interested in a standardized approach around crypto threat information? This is where some of the lessons came when we were standing up ISACs of cyber threat firms at first being like, "Well, this is our whole business model, sharing this info. How do we still monetize this?" Then they recognized that standards for that information to at least be machine-readable ended up not being an issue. They realized there was still a benefit to sharing that info, and that created a market for them to provide info in that format. I'm curious where we are in that evolution with the blockchain threat firms.

One thing that we're seeing increasingly is companies are subscribing to multiple blockchain analytics providers, and increasingly what they're really interested in is just getting the data from all of them and integrating it into their own systems. So, I think in that sense, it would be very useful for our customers. I can't really speak to the politics of that, but I think from a customer standpoint, it would be attractive.

I would assume that being able to create a highway where threats can just directly go where they're supposed to go will not impact the finding or the mining of that finding instruments, versus really just ensuring that someone actually looks at all the code or the logic of everything and tells me there is something wrong. But now, I want it to be almost instantly communicated to the interested parties, without having to find a new format or translate the data, or figuring out that there is a missing piece of data that could have been critical. Having the entire ecosystem look at these three pieces of data points makes a lot of sense.

To agree with what you said, it's like when we're looking at using the internet. We have the ISPs, the actual network providers, and they may be subject to different regulations. For example, someone posting a website to sell something physical may not be aware of how the telecom companies are actually regulated to ensure data goes where it's supposed to go. Similarly, in a situation with nodes and internal networks, someone hosting their own server to run a node for mining is very different from a huge data center. As an end-user, how do we ensure that the data we are transmitting is safe?

This ties into discussions on privacy and financial data sharing. In the U.S., we have the Cybersecurity Information Sharing Act, providing wide liability protection to any member of industry sharing cyber indicators, defensive measures, and threat information. In the financial domain, however, liability protection isn’t as extensive due to concerns about ensuring access to financial services. Financial institutions, under 314B provisions, can share transaction information to prevent money laundering, but it’s more restricted. Unlike cyber threat firms that

can share information broadly, blockchain analytics and LexisNexis can't operate with the same freedom. This leads us to questions about standards—like blacklists or gray lists of crypto wallets associated with illicit activities—and the implications for denying access to financial services based on these lists.

Another consideration is that Web3 is unique compared to the internet itself since it naturally deals with economic artifacts, money, shares, and other assets. This makes it more challenging, especially regarding standardization efforts. Has anyone estimated the effort needed to create such standards? Developing a crypto asset sharing standard, as Thomas mentioned, took 25 people over four years. With such resources required, is it even feasible to undertake this without firm commitments from stakeholders?

The role of standards in Web3 differs, as enforcement isn't always practical. For example, legitimate crypto asset creators could voluntarily agree to norms, like a "UL" stamp for safety, signaling reliability without top-down enforcement. In this scenario, standards serve more as guidance or assurance, allowing users to trust assets they interact with. Some communities, like block explorers, have already taken steps towards standards by tagging tokens with labels like "spam," which is a public good that benefits everyone.

In the U.S., we're rolling out cyber trust marks and potentially crypto trust marks, which could serve as decentralized trust indicators. But questions remain: at what point does this become a credit score or a credit bureau? If wallets can be tagged with confidence scores, it may also lead to privacy concerns, as over time, these tags could evolve from indicating scams to broader trustworthiness assessments. How do we address this without it devolving into an overly restrictive system?

I'm very curious to examine the best way to achieve the desired outcome while preserving privacy. When financial information is mixed with technical vulnerabilities, we need to ensure that we all understand terms like vulnerabilities and threats in the same way. For instance, a vulnerability in the financial system may refer to something related to anti-money laundering (AML), while in a technical system, it's often about identifying and patching weaknesses. Each context requires distinct remediation processes, so aligning terminology and processes is critical.

Hi, just a question. Are there any efforts to collaborate with blockchain developers? For example, Ethereum improvement proposals (EIPs) address standards differently, focusing on technical specifications rather than protections against damage. So, while standards are discussed, it's a different approach. Are there programs to engage these developers or key influencers who lead in this space?

Blockchain is typically developed by permissionless communities, which operate like the internet community. Some organizations like IETF create consensus mechanisms that are inclusive, allowing all stakeholders to participate. In contrast, standards organizations like ISO

involve national voting processes to adopt global standards. There are different agreement protocols, each suited to their ecosystem. In blockchain, some standards arise from smaller, more limited communities. BGIN was established to expand participation, ensuring broader consensus across diverse stakeholders in creating standards.

Moving to international cooperation, in previous sessions, we discussed ISAC (Information Sharing and Analysis Centers) models. How can BGIN help with crypto ISACs? One idea is for BGIN to act like ICANN in the DNS community, gathering stakeholders globally to discuss policies and publish guidance—let's say "documents" to avoid implying rigid standards. This approach could build a common language, like a taxonomy, so everyone speaks consistently when discussing issues or writing documents.

Each crypto ISAC may operate within its own regional or national context. BGIN can support their international collaboration, bridging gaps across the crypto ecosystem. One suggestion was to establish an ecosystem map to facilitate collaboration, identifying and connecting relevant organizations globally. An ecosystem map could serve as a low-effort, high-reward activity to encourage collaboration with crypto ISACs, ensuring visibility for those interested in similar initiatives.

Another important concept is the role of an "ecosystem orchestrator," which some projects use to align interests, motivations, incentives, and values across stakeholders. Orchestrators don't have control over individual stakeholders but instead create cohesion by aligning incentives. For BGIN, an orchestrator role could support stakeholders by building alignment, possibly even through a fee-based model if value is being provided to each stakeholder.

BGIN could also play a role in developing educational materials for nations lacking crypto ISACs. Such content could include technical knowledge sharing and practical guidelines for setting up information-sharing channels. Since many countries lack crypto ISACs, sharing knowledge internationally is key.

Crypto ISACs are not just a U.S.-based concept, as they are envisioned for global collaboration. Ideally, an international network of ISACs would support regional needs while remaining interconnected globally. Since crypto operates without borders, having ISACs that limit themselves to national borders could be challenging, though multiple regional ISACs may emerge over time.

For the roadmap, after Block 11, BGIN plans to create a cybersecurity working group to continue these discussions. We need to expand our participant base, bringing in individuals from countries outside the U.S. and Europe. The cybersecurity working group will work on crafting educational content for countries without crypto ISACs, as well as examining cybersecurity information-sharing alignments that were discussed in previous sessions. BGIN also plans to collaborate wit

h ISO to align standards, potentially bringing these to the international level once they're finalized.

BGIN recently held a Layer 2 meetup and engaged with the ISO community, building bridges with ISO's blockchain committee. Moving forward, we will continue to work with this community to refine our cybersecurity documentation and align with their standards. We invite all participants of Block 11 and members of BGIN to join the working group as we move into this next phase.

Lastly, BGIN will share information about this working group on our Discourse channel, inviting contributions and feedback on charter items. We encourage everyone to join the ongoing discussions, as these are critical to shaping the roadmap for international crypto ISAC collaboration.

Thank you very much for your time, and let's take a break now.